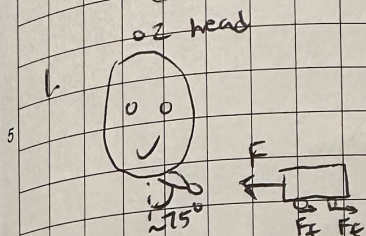


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Design requirements:



$$\Sigma F_x = F_{\text{total}} - F = 0$$

$$= \mu mg - F = 0$$

$$m = \frac{F}{\mu g}, \quad m = \frac{10}{0.8 \cdot 9.8} = 1.27 \text{ kg}$$

$$F = 10 \text{ N}$$

Robot weighs at least 1.27 kg to not get pulled back.

2. It takes 1.8 N to push all 3 levers at the same time. Assuming the contact between the robot and lever is 0.5s,

$$F \Delta t = mv, \quad v = \frac{F \Delta t}{m} = \frac{1.8 \cdot 0.5}{1.27} = 0.7 \text{ m/s}$$

Robot travel at least 0.7 m/s.

$$3. \quad \tau = Fr, \quad F = 1.8 \text{ N}, \quad r = 0.197 \text{ m}, \quad \tau = 0.35 \text{ Nm}$$

Robot must be able to generate at least 0.35 Nm to flip the levers any point from the pivot.

4. If we use a motor instead of the whole robot to flip the levers, the Power ~~needed~~ required ~~is~~ is

$$P = \tau \omega, \quad \omega = \frac{v}{r} = \frac{0.7 \text{ m/s}}{0.197 \text{ m}} = 3.55 \text{ rad/s}$$

$$P = 0.35 \cdot 3.55 = 0.888 \text{ watts}$$

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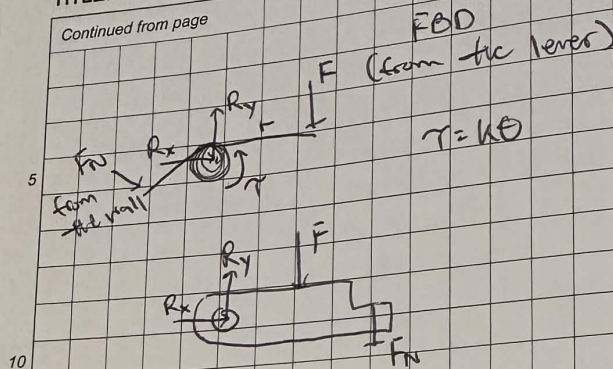
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PROPRIETARY INFORMATION

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Fabrication

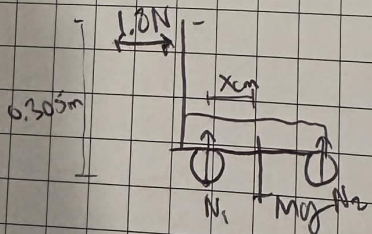
- I plan on using 1" aluminium square tubes to create the main part that looks like an E.
- Use aluminium brackets to attach the tubes
- Will use band saw to obtain
- Everything will be fastened by a screw and nut.

Design Reqs.

$$k\theta = rF, \quad k = \frac{rF}{\theta} = 0.0073 \text{ Nm}$$

$$r = 0.0127 \text{ m}, \quad F = 0.6 \text{ N}$$

$$\theta_{\max} = \frac{\pi}{3}$$

Roughly 0.0073 Nm
spring constant

$$1.8 \cdot 0.305 = X_{\text{cm}} m \cdot 9.8$$

$$m X_{\text{cm}} = 0.056 \text{ kgm}$$

The mass times X_{cm} must $\approx 0.056 \text{ kgm}$ to prevent flipping.

No numbers yet since X_{cm} depends on m .

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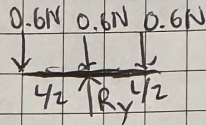
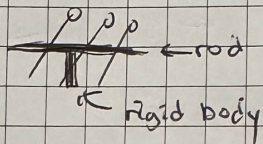
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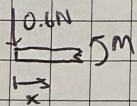
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5. If you push lever's head on:



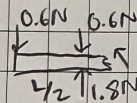
$$L = 0.254 \text{ m}$$

$$R_y = 1.8 \text{ N}$$



$$\sum M = M + 0.6x = 0$$

$$M = -0.6x$$



$$\sum M = M + 0.6x + (x - \frac{L}{2}) \cdot 0.6 - (x - \frac{L}{2}) \cdot 1.8 = 0$$

$$M = 0.6x - 0.6L$$

Max M at $\frac{L}{2}$, $M = -0.0762 \text{ Nm}$ Using a rod of radius 0.005 m , $I = \frac{\pi r^4}{4} = 4.91 \times 10^{-10} \text{ m}^4$

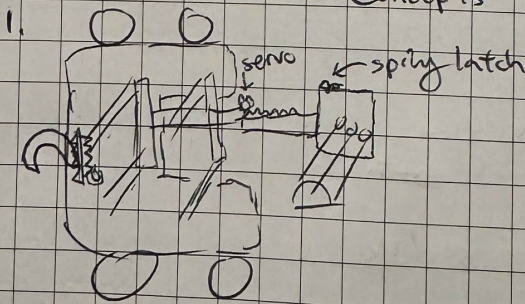
$$y_{\text{max}} = 0.005 \text{ m}$$

$$M = -0.0762 \text{ Nm}$$

$$I \sigma_{\text{max}} = \frac{M y}{I} = 776 \text{ kPa}$$

Basically don't use cardboard

Concepts



- Rack and pinion system
- Can be used to get machinists grimmerle
- Hook at the end to pull or head
- Another rack and pinion for the hook to grab the head

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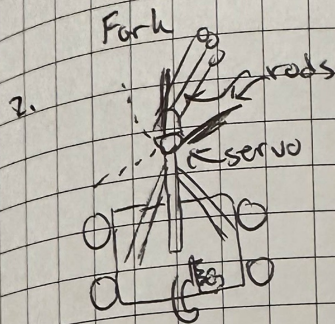
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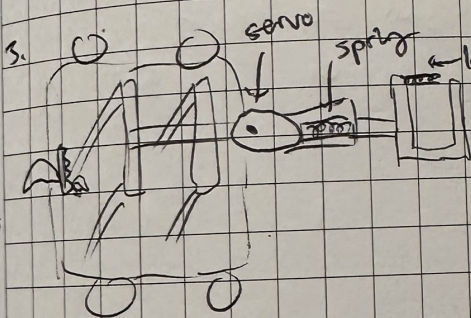
PROJECT:

BOOK:

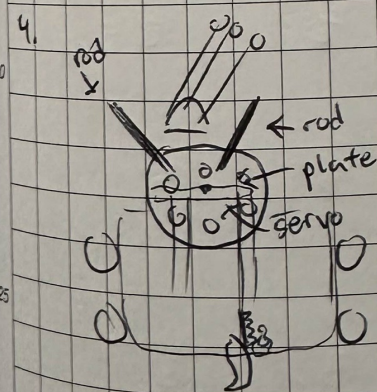
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- The servo rotates ~~cc~~ cc/cw and spins the 2 rods
- Hook at the end for the oz head



- Cam and follower system
- The spring allows the hook to return back to its original position
- If done correctly, I might be able to time the 1 second delay



- Instead of the fork that is parallel to the ground, the plate is perpendicular to the ground with the rods attached
- May be used to drive the train by switch sides of the plate
- Hook at the end

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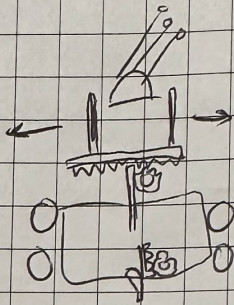
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5.



- Robot is stationary and only relies on the rods to move left and right

10

1 = excellent

2 = good

3 = decent

Concepts

1 2 3 4 5

15

Easy to set up to push levers?

2 1 2 1 1

Can push

Machinery's grimmerie?

1 2 1 2 2

20

Chance at fixing the 1s delay?

(based on markings, etc)

3 3 2 3 3

25

Ease of pushing lever (Based on pushing head on vs from the side)

1 3 1 3 3

30

total

7 9 6 9 9

Choose Concept 3

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